

Usage benefits	Example command
Synchronises directories	<code>rsync -avz /src/ /dest/</code>
Transfers files via SSH	<code>rsync -e ssh file user@host:/path</code>
Displays transfer progress	<code>rsync -av --progress /src/ /dest/</code>

Remote management

ssh: ssh (Secure Shell) is the standard tool for accessing and managing remote systems securely. It uses encryption to protect communications.

Usage benefits	Example command
Connects to remote servers	<code>ssh user@192.168.1.1</code>
Executes remote commands	<code>ssh user@host "command"</code>
Forwards local ports securely	<code>ssh -L 8080:localhost:80 user@host</code>

By leveraging these tools, administrators can maintain seamless operations while ensuring the security and efficiency of data transfers and remote access.

Network security tools

Network security tools are vital for protecting IT infrastructure from threats. They help administrators manage firewalls, detect intrusions, and mitigate potential risks. Proper use of these tools ensures a secure and resilient network environment.

Key questions administrators face daily with respect to network security are:

- Are firewalls configured to block unauthorised access effectively?
- Is the network monitored for signs of intrusion or malicious activity?

Firewall management

iptables: iptables is a powerful command-line tool for configuring packet filtering rules in Linux. It allows administrators to manage incoming and outgoing traffic effectively.

Usage benefits	Example command
Configures packet filtering rules	<code>sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT</code>
Lists current rules	<code>sudo iptables -L -v</code>
Deletes specific rules	<code>sudo iptables -D INPUT 1</code>

ufw: ufw (Uncomplicated Firewall) simplifies firewall configuration, making it user-friendly for administrators.

Usage benefits	Example command
Enables the firewall	<code>sudo ufw enable</code>
Allows specific ports	<code>sudo ufw allow 22/tcp</code>
Checks firewall status	<code>sudo ufw status verbose</code>

Intrusion detection

fail2ban: fail2ban monitors log files for suspicious activity and blocks IPs with malicious behaviour, such as repeated login attempts.

Usage benefits	Example command
Protects against brute force attacks	<code>sudo fail2ban-client status sshd</code>
Adds custom ban rules	<code>sudo fail2ban-client set sshd banip IP_ADDRESS</code>
Lists banned IPs	<code>sudo fail2ban-client banned</code>

suricata: suricata is an advanced intrusion detection and prevention system (IDS/IPS) that analyses network traffic for threats in real time.

Usage benefits	Example command
Runs in IDS mode	<code>suricata -c /etc/suricata/suricata.yaml -i eth0</code>
Logs detected threats	<code>cat /var/log/suricata/fast.log</code>
Updates rule sets	<code>suricata-update</code>

Network security tools like iptables, ufw, fail2ban, and suricata are vital for protecting IT infrastructure. They enable administrators to enforce access controls, detect threats, and respond to vulnerabilities, ensuring a secure and resilient network environment.

Performance and troubleshooting tools

Performance and troubleshooting tools are critical for maintaining an efficient network. These tools help administrators test bandwidth, diagnose latency, and identify network issues. By leveraging these tools, administrators can ensure high availability and optimal performance.

Key questions administrators face daily in this domain are:

- Is the network bandwidth sufficient to meet current demands?
- Are there latency issues or connectivity bottlenecks impacting performance?